Concepts in Statistics

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1. Variable Classifications

1.1 Continuous/Discrete distributions

Continuous probability distribution is a type of distribution that deals with continuous data or random variables:

- Probabilities are given for a **range of values**, rather than a particular value.
- For a continuous random variable (e.g., many biological measurements), the probability of any single value is **infinitesimally small**; however, the cumulative distribution function (cdf) can be easily computed.
- If we **differentiate** the cdf, we obtain the probability density function (pdf), denoted as $f(x)$, which describes a continuous variable. If it is discrete then it is probability mass function (pmf)
- The probability function accompanying a continuous random variable is a **continuous mathematical function**, represented as:

$$\int f(x)\,dx = 1$$

This integral ensures the total probability over all possible values equals 1.

1.2 Cumulative Distribution Function

The **cumulative distribution function (CDF)** adds all the values before x and shows the probability of having a value smaller than or equal to x:

$$F(x) = \sum_{x_i \leq x} P(X = x_i)$$

For a discrete random variable following the **uniform distribution**, the formula becomes:

$$F(x) = \frac{1}{6} \cdot n$$

Where n represents the number of values less than or equal to x in the uniform distribution.

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2. Probability Distributions & Probability Function

A probability distribution lists all the probability associated with the value of a random variable (x).

$P_{(x)}$ is a number from 0 to 1.

The area under a probability function is always 1.

Typical ones:

2.1 Uniform Distribution

- **Definition**: All outcomes within a specified range are equally likely.
- **Example**: Rolling a fair six-sided die: each side has a probability of $\frac{1}{6}$.
- **Maths**:
  - **PDF**: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$
  - **Mean**: $\frac{a+b}{2}$
  - **Variance**: $\frac{(b-a)^2}{12}$

2.2 Bernoulli Distribution

- **Definition**: A discrete distribution representing a single trial with two possible outcomes (success or failure), with a probability p of success.
- **Example**: Flipping a coin where $p = 0.5$ for heads and $1 - p = 0.5$ for tails.
- **Maths**:
  - **PMF**:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

  - **Mean**: p
  - **Variance**: $p(1 - p)$

2.3 Binomial distribution

- **Definition**: A discrete distribution representing the number of successes in n independent trials, each with a probability p of success.
- **Example**: Tossing a coin 10 times where $p = 0.5$, the number of heads follows a binomial distribution with $n = 10$ and $p = 0.5$.
- **Maths**:
  - **PMF**:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \ldots, n$$

  - **Mean**: np
  - **Variance**: $np(1 - p)$

2.4 Normal Distribution

- **Definition**: A continuous, symmetric distribution characterized by its bell-shaped curve, described by its mean (μ) and standard deviation (σ).
- **Example**: Heights of individuals often follow a normal distribution with a mean height of $\mu = 170\,\text{cm}$ and standard deviation $\sigma = 10\,\text{cm}$.

- **Maths**:
  - **PDF**:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  - **Mean**: μ
  - **Variance**: σ^2
- **Characteristics**: the 68-95-99.7 rule.

2.5 Standard Score (Z-score)

$$Z = \frac{X - \mu}{\sigma}$$

Where: - Z: The Z-score. - X: The raw score (data point). - μ: The mean of the population. - σ: The standard deviation of the population.

Z-score measures how many standard deviations a data point is from the mean (distance from the mean).

1. A **Z-score** of 0 indicates that the data point is exactly at the mean.
2. Positive Z-scores indicate values **above the mean**, while negative Z-scores indicate values **below the mean**.
3. Reported as "the data point is 2 standard deviations above the mean."

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3. Sampling

3.1 Sampling Methods

3.1.1 Single random sampling

Every member of the population has an equal chance of being selected.

3.1.2 Stratified sampling

The population is divided into strata based on certain characteristics, and random samples are taken from each strata according to strata size.

3.1.3 Two-stage cluster sampling

The population is divided into clusters groups. Some clusters are **randomly** selected, and then individuals within those selected clusters are randomly sampled. A cost-effective method for large populations spread over wide geographical areas.

3.2 Extreme Sampling

Select the most extreme or deviant cases (very high or very low) in a population.

Even if not representative, a sample can be useful (e.g. provide an upper/lower bound of sth)

3.3 Sampling Error (due to randomness, cannot be eliminated)

Definition: the difference between sample estimate and population measure.

All samples are wrong, some are more useful than others so we can:

- Colloect more samples
- Collect better samples

3.4 Sampling Bias (due to sampling method, can be eliminated)

Due to non-random selection of items, whether intentionally or not.

3.5 Sampling Distribution

Take samples of size n from a population which follows a for example, normal distribution (value-number) and record the *mean* for multiple times x.

The distribution of the sample mean is **sampling distribution** (mean-density).

- The *mean of the sample* in *sampling distribution* = the *value of the individual* in the *population distribution*.
- The *density of the mean* in *sampling distribution* = the *number of the value occurrence* in the *population distribution*.
- The more sampling time increases the more sampling distribution resembles the sample
- The more sample size increases the more towards-the-center the sampling distribution will get

3.5.1 Standard Error of the Mean (SEM)

A measure of how well the sample mean estimates the population mean.

$$SEM = \frac{\sigma}{\sqrt{n}}$$

- σ represents the standard deviation of the population.
- n is the sample size.

The more `n` increases the more towards-the-center the sampling distribution will get, the less SEM will be.

3.5.2 The Central Limit Theorem

Even if a population is not normally distributed, the sampling distribution for large enough sample sizes n will tend to be normal.

The more n increases the more towards-the-center the sampling distribution will get, the less SEM will be, the more normal will it be.

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4. Compare 2 Samples non-statistically

Depends on

1. Effect Size
2. Sample Size

This is the base of the rationale of t-statistic:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

4.1 Effect Size

A quantitative measure of the magnitude of the difference between groups. It helps to uncover the practical meaning of findings, beyond just statistical significance, but magnitude of the effect.

Cohen's d: Used to measure the standardized mean difference between two groups:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s}$$

- $\bar{X}_1$ and $\bar{X}_2$ are the means of the two groups. - s is the pooled standard deviation. - Small effect: $d = 0.2$ - Medium effect: $d = 0.5$ - Large effect: $d = 0.8$

4.2 Sample Size

n, directly affects SEM. It determines whether the comparison is valid/significant.

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5. Statistical Inference

Infer properties of the population from sampled data using statistical tests.

Is composed of **estimation** and **hypothesis testing**.

Choose the population property that we are interested in

Such as, "are the mean of these two populations different?", "is the variance of the population the same?", "is A takes up the same proportion in these two populations?"...

Estimation (hat) to Parameter (non-hat).

Estimate or Hypothesis Testing?
